

Database Identifier Normalization - Issue Resolution

Date: 2026-03-02

Status: Solution Designed

Component: TMS Bridge (Disposition-Abstraction-Layer)

Issue: Database identifier 10/1 not resolved correctly

Overview

This exploration addresses the issue where TMS Bridge fails to resolve company 10, branch 1 when provided as D-10-01 instead of D-10-1. The solution makes TMS Bridge resilient by normalizing database identifiers, ensuring both formats resolve to the same TMS database.

Files in This Exploration

1. [solution-design.md](#) - Comprehensive solution design document
 - Problem analysis
 - Implementation strategy
 - Configuration requirements
 - Testing approach
 - Migration path
2. [IMPLEMENTATION.md](#) - Step-by-step implementation guide
 - Exact code changes required
 - File locations
 - Testing checklist
 - Deployment strategy
 - Rollback plan
3. **BranchDbContextFactory-proposed.cs** - Complete proposed implementation
 - Ready-to-use code with normalization logic
 - Enhanced logging
 - Full documentation
4. **BranchDbContextFactoryTests-additions.cs** - Comprehensive test suite
 - Unit tests for normalization

- Integration tests for schema resolution
- Edge case coverage
- Tests for the specific issue reported

Problem Statement

Current Behavior

- D-10-1 → schema tms101 ✓
- D-10-01 → schema tms1001 ✗ (wrong, causes lookup failure)

Target Behavior

- D-10-1 → schema tms101 ✓
- D-10-01 → normalized to D-10-1 → schema tms101 ✓

Solution Summary

Add a **NormalizeDatabaseIdentifier** method that:

1. Extracts company and branch numbers from identifier
2. Parses them as integers (removes leading zeros)
3. Reconstructs identifier in normalized format
4. Uses normalized identifier for all lookups and cache keys

Key Changes

Single File Modified: BranchDbContextFactory.cs

- Add normalization method (~20 lines)
- Call normalization at entry point of CreateDbContext
- Enhanced logging for debugging

Test File Updated: BranchDbContextFactoryTests.cs

- Add 7 new test methods
- Cover normalization logic
- Verify issue resolution

Impact Analysis

Positive Impacts

- ✓ Resolves the 10/1 issue
- ✓ Prevents future similar issues
- ✓ Improves cache efficiency
- ✓ Backward compatible
- ✓ Clear error messages
- ✓ Self-documenting code

Risks

- ⚠ Low - Configuration keys might need updating to normalized format
- ⚠ Low - Minimal performance impact (~0.1ms per request)

Complexity

- **Low** - Single method, isolated change, comprehensive tests

Next Steps

1. **Review** - Team reviews solution design and implementation guide
2. **Approve** - Stakeholders approve the approach
3. **Implement** - Developer applies code changes
4. **Test** - Run unit and integration tests
5. **Deploy** - Roll out to dev → staging → production
6. **Verify** - Confirm both formats work in production
7. **Document** - Update API docs with normalization info

Related Issues

- OG reported 10/1 resolution failure
- Pascal Leicht questioned "01" vs 1 formatting
- Discussion about TMS numeric vs. string representation
- EBV rollout discovered the inconsistency

Technical Context

Database Identifier Pattern

`[D|0]-{company}-{branch}`

where:

D/0 = Database type prefix

company = 1-2 digit number

branch = 1-3 digit number

Schema Naming Convention

`{vendor_prefix}{company}{branch}`

Examples:

PostgreSQL: tms101, tms1034, tms2820

Oracle: TMS101, TMS1034, TMS2820

Current Regex

```
@'^[D0]-(\d{1,2})-(\d{1,3})$'
```

Accepts both formats but doesn't normalize them.

References

Code Locations

- **BranchDbContextFactory.cs**: Line 22-82
- **DbConnectionStringProvider.cs**: Line 15-24
- **GetDriversQueryHandler.cs**: Line 45-48 (shows usage in Backend)

Configuration Examples

- Code/Disposition-Backend/CALConsult.Disposition.API/Resources/branchdescriptions.json
- Code/Disposition-Abstraction-Layer/CALConsult.TMSBridge.API/appsettings.json

Test Data

- Test cases use D-10-52 , D-20-200 , D-10-34
- Backend tests use D-10-1 , D-1-1 , D-28-20

Approval Required From

- ☐ Lead Developer (code review)
- ☐ TMS Bridge Team (solution validation)
- ☐ New Dispo Backend Team (integration confirmation)
- ☐ DevOps (deployment planning)

Time Estimate

- **Implementation:** 2 hours
- **Testing:** 2 hours
- **Code Review:** 1 hour
- **Deployment:** 1 hour
- **Total:** ~6 hours

Success Metrics

After deployment, verify:

1. Both D-10-1 and D-10-01 work in API calls
2. No errors in logs related to database resolution
3. Cache hit rate maintains or improves
4. No performance degradation
5. All existing functionality continues to work

Conclusion: This is a well-scoped, low-risk solution that addresses the root cause of the database identifier resolution issue while improving the overall resilience of the TMS Bridge system.